

E-COMMERCE CONVERSION INSIGHTS

Data Science & Analytics Internship – Task 3

Submitted By:

Devisri Chowdary Koya

1. Introduction

The E-Commerce Conversion Insights project focuses on analyzing customer engagement, purchasing behavior, and conversion performance in an e-commerce platform using Power BI and Python.

The primary objective of this project is to understand how customers interact with products through activities such as product views, cart additions, and purchases in order to evaluate customer behavior and business performance.

Interactive dashboards and analytical visualizations were developed to generate meaningful insights that support data-driven business decisions.

2. Objectives

- Analyze customer conversion behavior
 - Monitor purchase activity and engagement trends
 - Identify top-performing brands based on revenue
 - Visualize category-wise customer purchases
 - Develop interactive dashboards using Power BI
 - Perform additional analysis using Python in Jupyter Notebook
-

3. Tools & Technologies Used

- Power BI
- Python
- Pandas
- Matplotlib
- Seaborn

- Jupyter Notebook
-

4. Dashboard Features

The Power BI dashboard includes:

KPI Metrics

- Total Views
- Total Cart Additions
- Total Purchases
- Total Revenue

Interactive Visualizations

- Conversion Funnel Analysis
- Purchase Trend Analysis
- Top Brands by Revenue
- Category-wise Purchase Distribution
- Customer Engagement Overview

Interactive Filters

- Event Time
- Product Category
- Brand

The dashboard delivers an interactive and professional analytics experience for tracking customer activity and overall business performance.

5. Python Analysis

Additional analysis was performed using Python in Jupyter Notebook.

The following visualizations and analyses were created:

- Customer Activity Distribution
- Revenue Analysis by Brand
- Purchase Trend Visualization

- Top Product Categories
- Customer Engagement Share
- Top Active Users Analysis

Python libraries such as Pandas, Matplotlib, and Seaborn were used for data cleaning, preprocessing, analysis, and visualization.

6. Key Insights

- Product views were significantly higher compared to completed purchases.
 - Electronics and smartphone-related categories attracted higher customer engagement.
 - Certain brands generated noticeably higher revenue than others.
 - Customer activity remained active throughout the analyzed time period.
 - The conversion funnel highlighted customer drop-offs between cart additions and purchases.
-

7. Conclusion

This project successfully demonstrates the application of Power BI and Python for customer behavior analysis and business intelligence reporting.

The dashboard provides clear insights into customer interactions, conversion performance, purchasing behavior, and revenue trends through interactive visualizations and analytical reporting.

This project enhanced practical skills in:

- Data Visualization
- Dashboard Development
- Business Analytics
- Python Programming
- Power BI Reporting
- Customer Behavior Analysis

The final dashboard and analysis provide a professional and interactive business analytics solution suitable for internship projects, portfolio development, and real-world reporting applications.

